

Multi-method geophysical approach to assess the internal porous structure of a deep-seated slow-moving landslide

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ARTICLE INFO

Keywords:

Gravimetry
Electrical resistivity tomography
Near-surface geophysics
Landslide
Natural hazards

ABSTRACT

Imaging the internal structure of landslides and characterizing their dynamics and related hazards remains a significant challenge. This study investigates the internal structure of a slow-moving rainfall-induced landslide in southern France. The region is particularly susceptible to landslides, with many posing threats to local infrastructure. For this study, we select a typical landslide that is representative of local instabilities, easily accessible, and already instrumented. This landslide has damaged a bridge and threatens the nearby road and several buildings. We used an integrated approach that combines electrical resistivity measurements, microgravimetric data, morphological and geological observations, and borehole information to infer the landslide's internal structure. Our findings reveal the presence of a negative density-high electrical resistivity anomaly, which correlates with colluvial deposits observed at the surface and in borehole cuttings. Along the studied profile, this body covers most of the bedrock layers, spanning over 700 m with an average thickness of 30 m. It exhibits significant thickness variations (15–45 m), attributed to a complex colluvium-bedrock interface geometry due to tilted limestone blocks. Its effective porosity, derived from the gravity survey, is 0.23 ± 0.05 , indicating a high storage capacity and significant pore-water pressure variations caused by heavy rainfall, which can lead to slope movement. This study highlights the complementarity of gravimetry with conventional geophysical methods, such as electrical resistivity tomography or borehole logs. Gravimetry offers more direct insights into rock density and associated effective porosity, which are essential for understanding the internal dynamics of landslides.

1. Introduction

A better understanding of the impact of precipitation on landslide stability is essential for improving hazard assessment and implementing effective risk mitigation strategies. This issue is made all the more acute because, in the context of climate change, heavy precipitation episodes are becoming more intense and frequent (e.g., Zhao et al., 2025; Araújo et al., 2022; Gariano and Guzzetti, 2022; Bernardie et al., 2021). Although the correlation between rainfall and landslide dynamics is now well established (e.g., Segoni et al., 2018), the spatial and temporal changes in water seepage, groundwater storage linked to effective porosity, and flow that lead to pressure variations and subsequent sliding remain poorly documented.

To tackle this issue, the first step is to characterize the internal

structure of landslides, particularly in terms of geometry and hydrological properties. Over the past few decades, numerous geophysical methods have been used (e.g., Pazzi et al., 2019; Whiteley et al., 2019) and combined to adopt multi-method approaches to enhance resolution, reduce ambiguities due to non-uniqueness of models, and improve the reliability of the geophysical images (e.g., Ciampi et al., 2025; Lu et al., 2024; Wróbel et al., 2023). Electrical resistivity tomography (ERT) and seismic refraction tomography (SRT) are the most common geophysical techniques used for landslide imaging (in 2D, 2.5D, and 3D) in stand-alone profiles or time-lapse measurements (e.g., Sun et al., 2024; Whiteley et al., 2021; Ehrlich et al., 2018). Other geophysical techniques, including various seismic methods (reflection, ambient noise, surface waves) (e.g., Provenzano et al., 2025; Zhang et al., 2025), self-potential (e.g., Hu et al., 2024) and induced polarization methods (e.

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<https://doi.org/10.1016/j.jappgeo.2026.106485>

Received 17 April 2025; Received in revised form 5 May 2026; Accepted 20 August 2026

Available online 24 August 2026

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g., Souza de Araújo et al., 2025) are used: (i) to improve the resolution and the depth of investigation of the geophysical images, (ii) to detect precursor signals before a landslide failure, and (iii) to map permeability, water content, and clay content. Additional geotechnical and borehole data are sometimes included in the multi-method approach to constrain or validate geophysical investigations (e.g., Bièvre and Crouzet, 2021).

Microgravity measurements, while rarely used to characterize landslides, offer complementary sensitivity to variations in electrical and velocity contrasts. One can mention the pioneering study of Del Gaudio et al. (2000), which demonstrated the effectiveness of microgravimetric data in assessing the internal structures of landslides and revealed that coarse colluvial fillings can explain the observed Bouguer anomaly. Additionally, several studies have highlighted the benefits of using microgravity measurements to reduce the interpretational ambiguities that can arise from relying solely on single-method approaches, such as ERT for imaging failure surfaces or faults within and beneath the sliding mass (Sastry and Mondal, 2013; Sastry et al., 2008) or bedrock depth (Táborík et al., 2017).

Our study focused on the Pégairolles-de-l'Escalette landslide, a slow-moving, rainfall-induced rotational landslide covered by colluvial deposits that may significantly influence rainwater infiltration and circulation within the hillslope. We used an approach that coupled electrical resistivity measurements and microgravimetry acquisitions to produce the first geophysical image of its internal structure. Even if the ERT has proven to be a powerful tool for inferring the hydrological state of a hillslope, variations of the electrical resistivity can be influenced by geological or pore-fluid properties such as saturation or salinity (Waxman and Smits, 1968; Watlet et al., 2024). In such cases, microgravimetry is a valuable complementary tool, providing independent information on density variations and, consequently, on the hillslope's internal porous structure (e.g., Chaffaut et al., 2022).

The main objective of this study is to introduce a novel multi-method approach that integrates electrical resistivity tomography (ERT), microgravimetry, and borehole data to characterize the internal porous structure of a deep-seated, slow-moving landslide. We specifically aim to demonstrate the value and advantages of a coupled geophysical approach, with particular emphasis on the contribution of

microgravimetry still rarely applied in landslide investigation, for imaging landslide internal structures. After a brief overview of the Pégairolles-de-l'Escalette landslide, including its geological setting and available information, we describe the geophysical fieldwork conducted at this site. Next, we present and interpret the geophysical images, integrating geological and borehole observations to outline the internal structure of the slope. Finally, we analyze the spatial variation in colluvial deposit thickness and evaluate the associated effective porosity. A comparison with the pore-pressure borehole data underscores the key role of slope deposits in landslide dynamics.

2. Study area

The Pégairolles-de-l'Escalette landslide is located in southern France, at the edge of the Larzac karst plateau, between the Cévennes mountains and the Mediterranean Sea. It lies along the western bank of the Lergue River in the Lodève basin (Séranne et al., 2002) (Fig. 1). The geological bedrock along the Lergue valley comprises thick and massive Jurassic carbonate units covering upper Triassic evaporite layers (Alabouvette et al., 1988). In this area, the Jurassic layers are dominated by the Hettangian dolomudstones, including the mudstones of the Parlatges facies at their base. The lithological succession of the upper Triassic sequence encompasses dolomitized grainstones and mixed siliciclastic carbonate deposits of the Rhaetian unit and alternating layers of white gypsum/anhydrite and claystone of the deeper Carnian-Norian units (see Schneider et al. (2006) and Lopez et al. (2005) for a detailed description of the playa environment in the Lodève basin).

The incision of the Larzac plateau by the Lergue River through the tabular middle to lower Jurassic units down to the Carnian-Norian sequences shapes the morphology of the valley and initiates slope instabilities (Fig. 1) (<https://www.georisques.gouv.fr/donnees/bases-de-donnees/base-de-donnees-mouvements-de-terrain>). Erosion mechanisms involve gravitational processes from rock falls and toppling (Gasc-Barbier et al., 2024; Taboada et al., 2017) to mass movement mechanisms further down the valley, where various scales of rotational landslides occur within weaker marl-limestone layers (Elkharat et al., 2024a). Geomorphological features suggest that deep rupture surfaces rooted in the Triassic are responsible for the rotational slides (Hungr

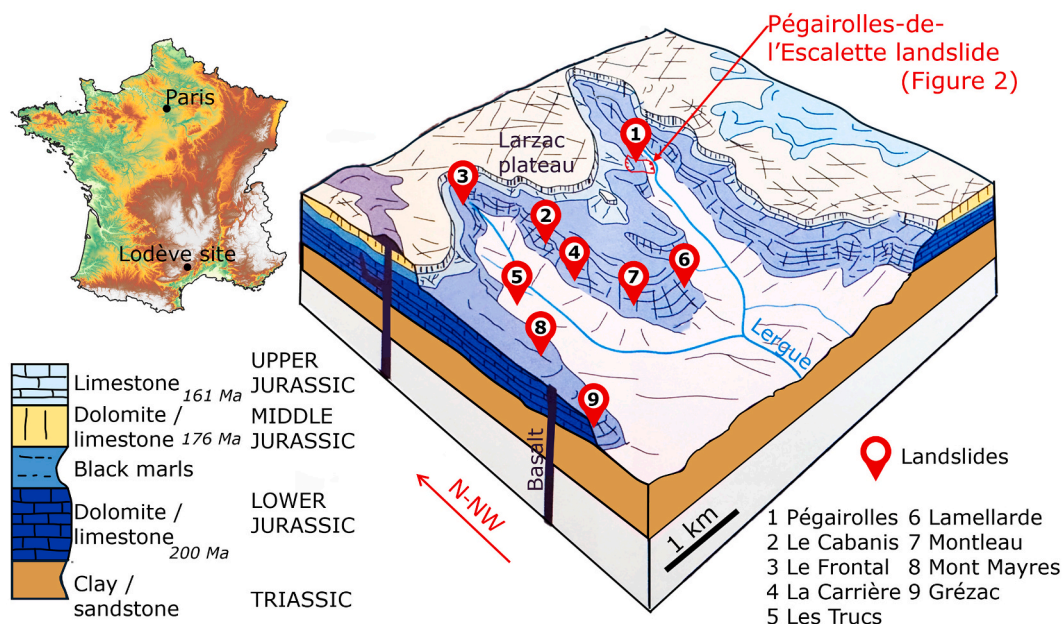


Fig. 1. Diagram showing the Lodève study site and the Pégairolles-de-l'Escalette landslide among the instabilities (red pins) observed in the Lergue valley in southern France. A simplified rock column illustrates the main lithology and age of rock units forming the Larzac plateau. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

et al., 2014; Gutiérrez et al., 2012).

Landslide hazards in this area are primarily triggered by short-duration heavy rainfall events, which can yield between 200 and 400 mm of rain over 1 to 3 days (Devers et al., 2021; Blanchet and Creutin, 2017; Delrieu et al., 2005). These events typically follow dry summer periods, when warm, humid air masses from the Mediterranean Sea converge with cold, high-altitude air masses at the southern edge of the Cévennes mountains. This region experienced multiple tectonic episodes that created a series of vertical north-south faults and fractures that today favor vertical seepage of meteoric waters at depth down to the deep Triassic layers, inducing ductile creep behavior and slope deformation (Elkharat et al., 2024a; Gutiérrez et al., 2012). Although the correlation between precipitation and deformation has been demonstrated for many landslides (e.g., Guzzetti et al., 2022; Polemio and Petrucci, 2000), the feedback between water infiltration, groundwater flow, and slope deformation processes remains poorly constrained. This prevents a relevant hazard assessment in this area, where landslides

damage numerous infrastructures such as the A75 highway, bridges, and houses.

The Pégairolles-de-l'Escalette landslide is representative of those in the Lergue valley. It is a typical rotational landslide that has been active from the Late Pleistocene to the Holocene (Elkharat et al., 2024b). This landslide is 750 m long and 650 m wide on the surface (Fig. 2), with a moving deformation of 3–4 mm/year (Denchik et al., 2019). This slow movement facilitated the deployment of an original long-term near-surface and downhole monitoring as part of the French National Landslide Observatory (SNO OMIV) (Fig. 2). PEG1-GM, DEO (Downhole electrical observatory), WB (West-Bay) are 60 m-depth boreholes drilled in 2012. Among these, the DEO borehole is cored. DEO and WB boreholes are equipped with permanent instrumentation, allowing temporal monitoring of electrical resistivity and hydrogeology at depth, respectively. PEG1-BRGM, drilled in the 1960s to a depth of 320 m, provides an additional lithological log on the southern slope (<http://ficheinfoterre.brgm.fr/InfoterreFiche/ficheBss.action?id=BSS002EPZF>).

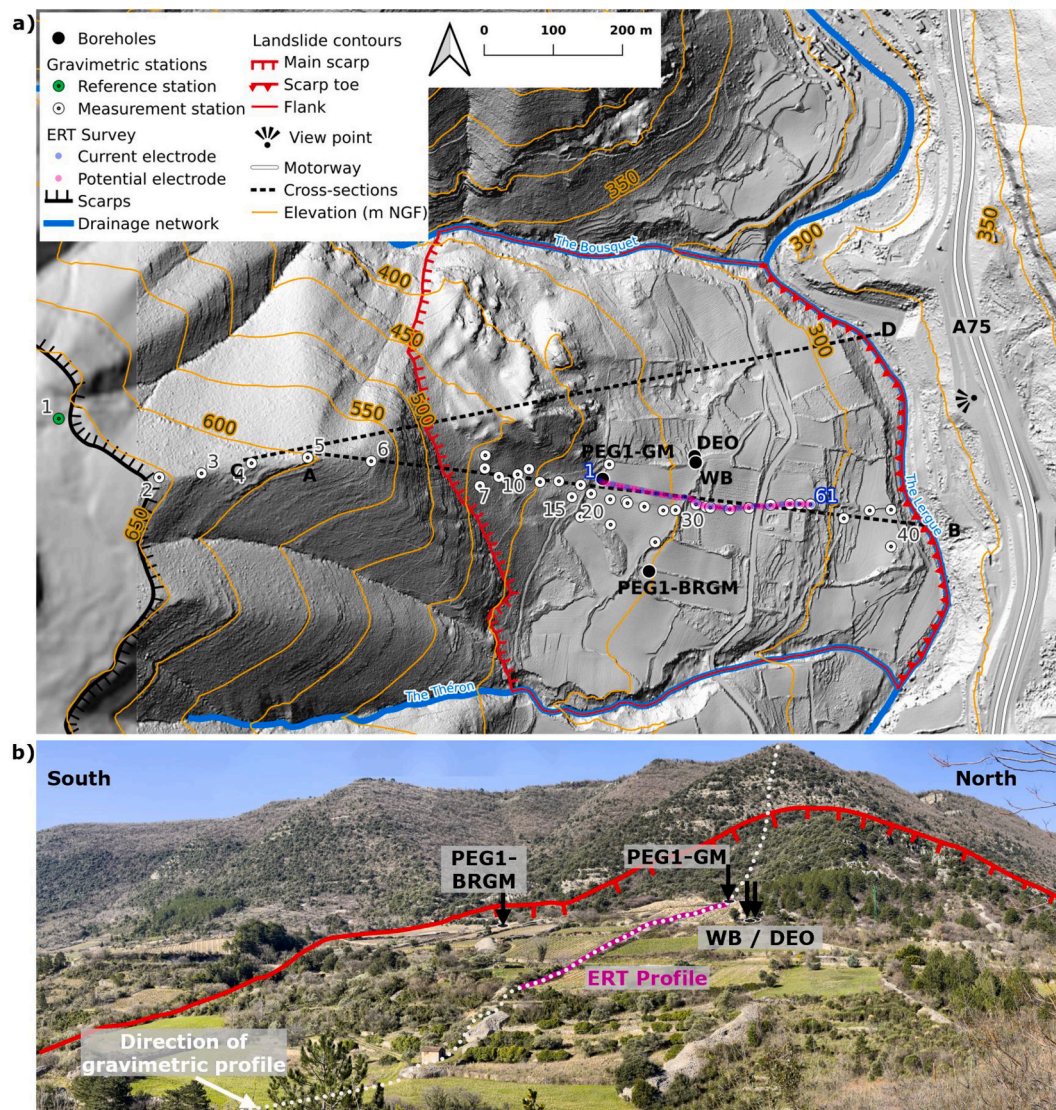


Fig. 2. Presentation of the study site. a) Shaded relief map of the Pégairolles-de-l'Escalette landslide showing the main geomorphic features, along with the location of boreholes (black circles and corresponding names), microgravimetric measurements (black-and-white circles and associated station numbers) and electrical resistivity profile (pink line). The red lines indicate the landslide's boundaries. The orange lines show contour lines at elevations between 300 m and 650 m. The two black-dashed lines labelled A-B and C-D correspond to the studied profile and the cross-section proposed by Denchik et al. (2019), respectively. b) South-North panoramic photograph of the landslide taken on March 18, 2024, at the viewpoint indicated in Fig. 2a. Black arrows and associated names mark the locations of the four boreholes. The dotted white line and the solid pink line show the positions and directions of geophysical profiles. Red lines indicate the landslide's boundaries. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Denchik et al. (2019) describe the lithology and hydrogeology of this site based on conventional borehole data and downhole geophysical and hydrochemical records. During the drilling, cuttings showed that the first 32 m consisted of slope colluvial deposits. Logs and core description from the DEO borehole, available from depths of 32 m (Fig. 3), indicate that the landslide body beneath the colluvial deposits is composed of materials from three different lithological units, with a predominance of marly dolomite and limestone down to a depth of about 52 m and the massive Triassic evaporite-clay layer underneath. Using the triple weighing method, laboratory measurements on seven plugs from the DEO borehole yield a matrix density of $2700 \pm 100 \text{ kg.m}^{-3}$ for the dolomite and limestone. Breccia, clayey gouge, a high concentration of open discontinuities, and water circulations mark a failure active zone at around 45 m depth. This deformation zone delimits two aquifers within the unstable mass: a semi-confined reservoir in the lower evaporite-clayey unit and a fractured, karstic unconfined aquifer in the overlying Jurassic carbonates. PEG1-GM and PEG1-BRGM boreholes show spatial variations in the depth of the evaporite-clay interface, with a significant westward deepening. Finally, monitoring of DEO and WB boreholes reveals a seasonal response of the landslide with a two-step response to intense rainfall events: instantaneous pressure transfer and water mass transfer delayed by 3 months. This two-step response may originate from the failure zone, which constitutes a heterogeneity at the slope scale, affecting water infiltration, groundwater flow, and drainage towards the deeper part of the slope.

3. Field work and geophysical data acquisition

Borehole data, logs, and downhole monitoring provide valuable insights into the subsurface lithologies and the failure zone at depth (Fig. 3). However, these data are inherently limited to the borehole's location and provide no information on lateral variations, which are essential for delineating the internal geometry of the landslide and the associated groundwater flow pattern. To address this limitation and produce the first image of the internal structure, we complement existing borehole datasets by integrating ERT and microgravity measurements along a West-East profile located at the center of the Pégairolles-de-l'Escalette landslide (along the A-B direction on Fig. 2a). We acquired the data over several years between 2018 and 2024, taking care to

collect the measurements during the dry season. This approach allows us to accurately compare the observations, since they are unaffected by potential variations in electrical resistivity or density caused by rainfall events and associated groundwater storage or flow.

3.1. Electrical resistivity survey

Electrical resistivity tomography is a conventional geophysical method that images the spatial distribution of the resistivity in the subsurface by injecting electrical current into the ground and measuring the resulting potential differences. Different quadrupoles, with varying locations and spacings, are used to map lateral and deep variations, producing a 2D resistivity distribution map. Since electrical resistivity is related to factors such as lithology, porosity, and water content, ERT is particularly well-suited for investigating heterogeneous porous media. In landslide studies, ERT is widely used to map internal structures and delineate zones with contrasting water content (e.g. Perrone et al., 2014).

Since 2018, a permanent Geomond4D geoelectric system (Supper et al., 2009) has been installed in the lowest part of the landslide, buried at a depth of 60 cm (Fig. 2). The system comprises a 324-m-long geo-electrical device with 61 electrodes spaced 5.4 m apart. Among them, 16 electrodes are designated for current injection, each separated by three potential electrodes. The multi-gradient array is used for data acquisition to leverage the equipment's multichannel capability, effectively balancing acquisition time and resolution while maintaining a good signal-to-noise ratio (Loke, 2025; Dahlin and Zhou, 2006). This array is also well-adapted for accurately imaging complex media (Gance et al., 2016). This geoelectric device has collected 51 datasets consisting of 3050 measurements from October 24, 2018, to November 21, 2019. The datasets are available in raw and filtered formats on the OMIV (Observatoire Multi-disciplinaire des Instabilités de Versants) online data repository (<https://data.oreme.org/observation/omiv>). To ensure consistency with the weather conditions of gravimetric campaigns, we used the dataset acquired on March 21, 2019. The OMIV filtration procedures first remove duplicates and eliminate data with voltage outside the range $[-9.95; +9.95] \text{ V}$ or with current injection below 0.5 mA, thereby eliminating data points that are erroneous due to technical limitations of the monitoring system. Resistance measurements with

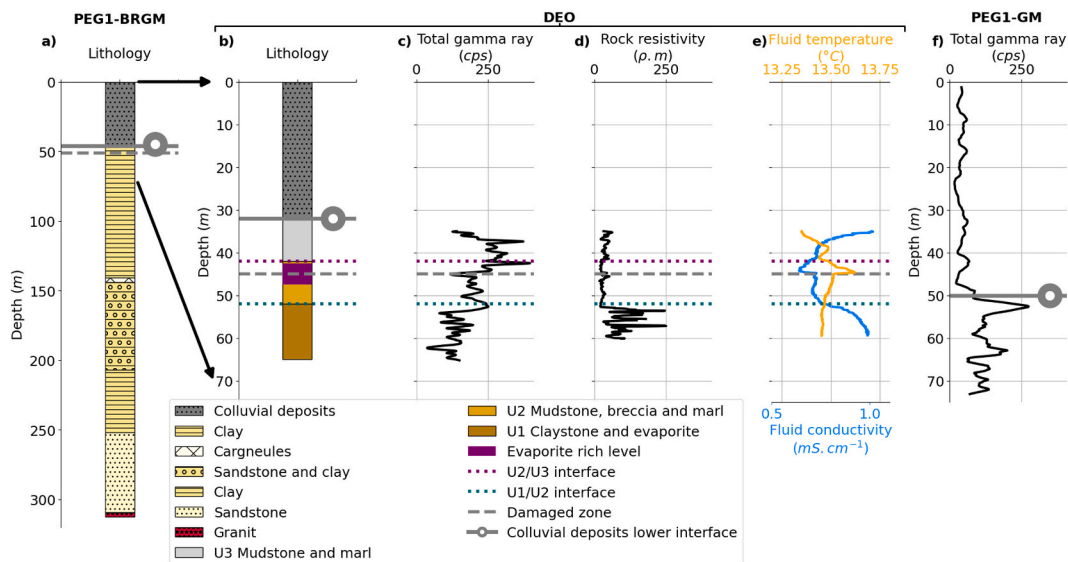


Fig. 3. Summary of the dataset from the three boreholes (PEG1-BRGM, DEO and PEG1-GM) located in the center of the Pégairolles-de-l'Escalette landslide. For the DEO borehole, we extracted lithological interfaces (represented by horizontal lines) and geological units (U1, U2, U3) as described by Denchik et al. (2019). Note the difference in the depth of investigation between the PEG1-BRGM borehole and the DEO and PEG1-GM boreholes. a) Lithological log for the PEG1-BRGM borehole. b) Lithological log for the DEO borehole. c) Total gamma ray in the DEO borehole. d) Electrical resistivity of rock in the DEO borehole. e) Fluid temperature and electrical conductivity in the DEO borehole. f) Total gamma ray data for the PEG1-GM borehole.

negative values and forward and reverse voltage measurements with the same sign are both eliminated, resulting in a filtered dataset of 3030 measurements. To remove inconsistent measurements, we keep only data within 3 times the median of the apparent resistivity distribution. Finally, the last criterion we used to eliminate the remaining outliers corresponds to a high data misfit ($> 100\%$) between observed apparent resistivity and the model response after a first independent inversion of the dataset. The resulting data collection includes 3008 measurements ranging from 132 to 2460 $\Omega\cdot\text{m}$, with a median of 668 $\Omega\cdot\text{m}$ (Fig. 4). The standard deviation of the electrical apparent resistivity repeated measurement by the resistivity meter is 0.05 $\Omega\cdot\text{m}$ on average.

3.2. Gravimetric campaigns

Microgravimetry is a geophysical method that uses small spatial variations in the Earth's gravitational field to image density contrasts related to the internal structure of massive bodies. Although gravimetric surveys offer dense coverage, they are seldom employed in landslide investigations due to the strong nonuniqueness of gravity inversion. A sufficient density contrast is essential for differentiating local effects caused by heterogeneities within the sliding mass from regional effects arising from larger-scale structures. However, Del Gaudio et al. (2000) showed that microgravimetric surveys, when supported by borehole measurements, have sufficient sensitivity to detect lithological heterogeneities within landslides. Here, for the first time, we proposed integrating microgravimetric measurements with other geophysical and borehole data to investigate the internal porous structure of the study landslide.

We collected gravimetric data along an approximately 1200-m-long profile aligned with the ERT measurements. This profile crosses the entire Pégairolles-de-l'Escalette landslide, extending from the Lergue River's western bank to the Larzac plateau's southern edge (Fig. 2). We performed microgravity measurements using Scintrex CG-5 and CG-6 gravimeters during three campaigns: June 2022 (18 measurements), June 2023 (19 measurements), and March 2024 (38 measurements). Stations 1 to 6, located at the western end of the profile on the Larzac plateau, are spaced about 100 m apart. Stations 7 to 40, located on the landslide, have a closer spacing of less than 20 m (Figs. 2 and 5). To

ensure that these relative gravity measurements could be adjusted to a common reference, each acquisition day began and ended with a series of measurements taken at the absolute base station located at the Géosciences Montpellier laboratory. Given that this reference point is approximately 60 km from the study area and, therefore, potentially affected by variations in instrumental drift between measurement campaigns, we reinforced the consistency between the different acquisitions by carrying out repeated measurements at gravity stations 17, 21, 31, 34, and 40, located in the eastern part of the measured profile.

We processed the raw gravity data with the GravProcess software (Cattin et al., 2015). We have not accounted for any effects of tide and pressure. We applied corrections for instrumental drift. We calculated the topographic effect from a 50 cm-resolution digital elevation model (DEM) provided by the Institut national de l'information Géographique et forestière (<https://geoservices.ign.fr/lidarhd>), assuming a standard density of 2670 kg m^{-3} . The quality of this DEM, derived from LiDAR data, offers altimetry accuracy within 10 cm. This accuracy is consistent with levelling measurements conducted on the Pégairolles-de-l'Escalette landslide, indicating average and maximum elevation errors of 0.06 m and 0.16 m, respectively. We assessed the accumulated error from instrumental accuracy, network adjustments, and corrections, which is less than 0.1 mGal for the complete Bouguer anomalies at the 40 gravity stations along the study profile (Fig. 5). This estimated error is supported by results from repeated measurements taken between 2022 and 2024, which displayed deviations of ca. 0.05 mGal.

The calculated anomalies exhibit a clear signal characterized by two primary wavelengths. The first wavelength encompasses the entire profile and features two zones with nearly constant Bouguer anomalies, separated by a sudden decrease of approximately -0.75 mGal located at the landslide main scarp, between -300 m and -200 m from the westernmost electrode (Fig. 5). The second wavelength, observed on the main body of the landslide in the eastern part of the profile, is shorter, measuring about 200 m. Its amplitude is associated with slight variations in the Bouguer anomaly, ranging from -0.9 to -0.55 mGal relative to the plateau values. Additionally, it is noteworthy that the values increase to -0.6 mGal in the easternmost part of the gravity profile, near the Lergue River (Fig. 5).

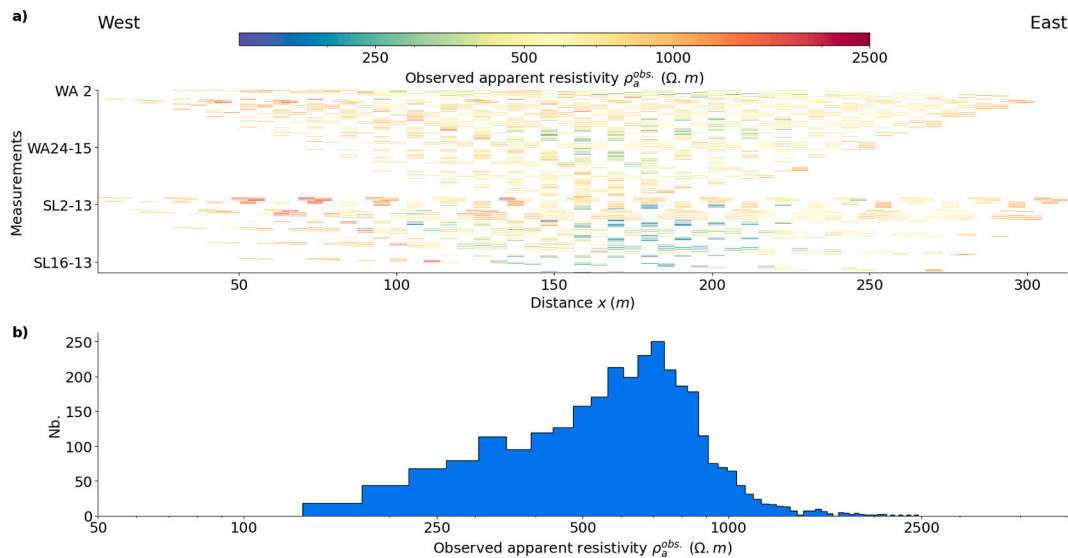


Fig. 4. Observed electrical apparent resistivity data ρ_a^{obs} collected with a multiple-gradient array along the West-East ERT profile shown in Fig. 2. The filtered data set contains 3008 values. a) Pseudo-section representation with a logarithmic scale colorbar. Data are not presented in a classical way due to the multiple-gradient array (Loke, 2025). The pseudo-section is displayed using the PyGIMLI toolbox (Rücker et al., 2017). The nomenclature used for the y-axis tick labels follows the “AAdd-Sep” format, where “AA” is the array type (“Wa” for “Wenner alpha” and “Sl” for “Schlumberger or Gradient”, respectively), “dd” is the potential dipole length and “sep” is the source-receiver distance, both of which are expressed in terms of the number of electrodes. For all cross-sections displayed in this study, the distance $x = 0$ m corresponds to the westernmost electrode of the electrical resistivity profile. b) Distribution of apparent resistivity values.

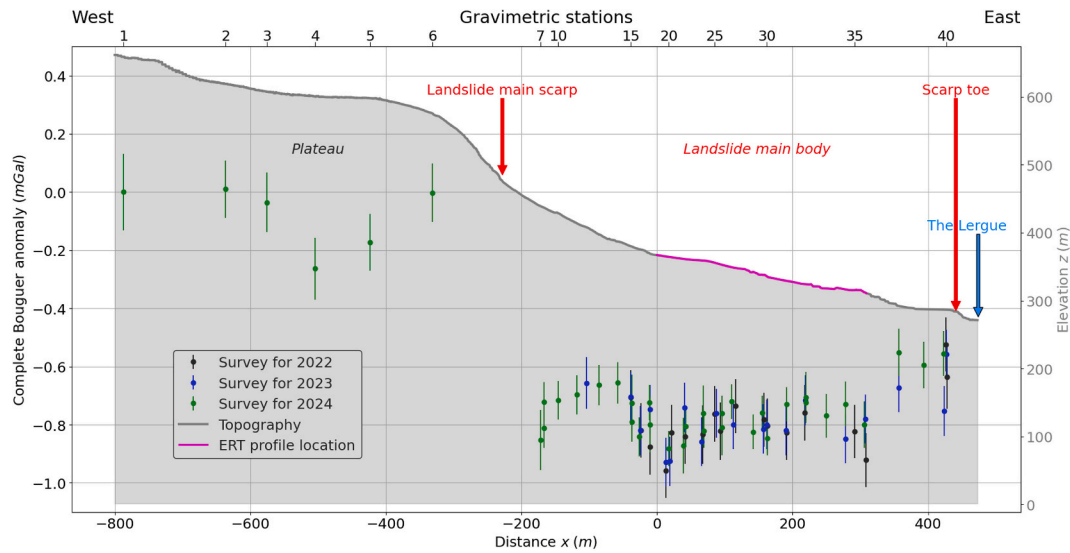


Fig. 5. Complete Bouguer anomaly relative to the measurement taken at the westernmost gravimetric station 1, located on the Larzac plateau. The colors of the circles correspond to the microgravimetric surveys conducted between 2022 and 2024. Red arrows indicate the landslide boundaries. The grey line shows the associated elevation profile extracted from the high-resolution digital elevation model provided by the Institut national de l'information Géographique et forestière. The section marked by the pink line indicates the location of the electrical resistivity profile. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4. Results

Geophysical data, including apparent resistivity and complete Bouguer anomaly, are processed using the open-source Python suite PyGIMLi (Rücker et al., 2017), which is specially designed for imaging subsurface structures. We used this software to invert the resistivity data and model the density contrast.

4.1. Electrical resistivity tomography

The filtered apparent resistivity dataset (Fig. 4) is inverted on an unstructured mesh of 8788 triangular cells. This mesh extends

approximately 100 m in both directions from the ends of the ERT profile and below the lowest electrode to minimize edge effects. The mesh is refined near the electrodes to accurately resolve the equations governing current fluxes and to better identify resistivity contrasts. The inversion uses a classical regularized Gauss-Newton scheme (Rücker et al., 2017; Günther et al., 2006; Park and Van, 1991). The regularization factor is set to $\lambda = 20$, the error value used in chi-squared calculation is 10%, and the initial homogeneous resistivity model is based on the median value of the data, which is 668 $\Omega \cdot m$. After five iterations, the chi-squared value decreased from 18.6 to 2.2 (a 88% reduction), and the relative root mean square (RMS) error from 37% to 17%, indicating a significant reduction in misfit. The comparison between observed and simulated

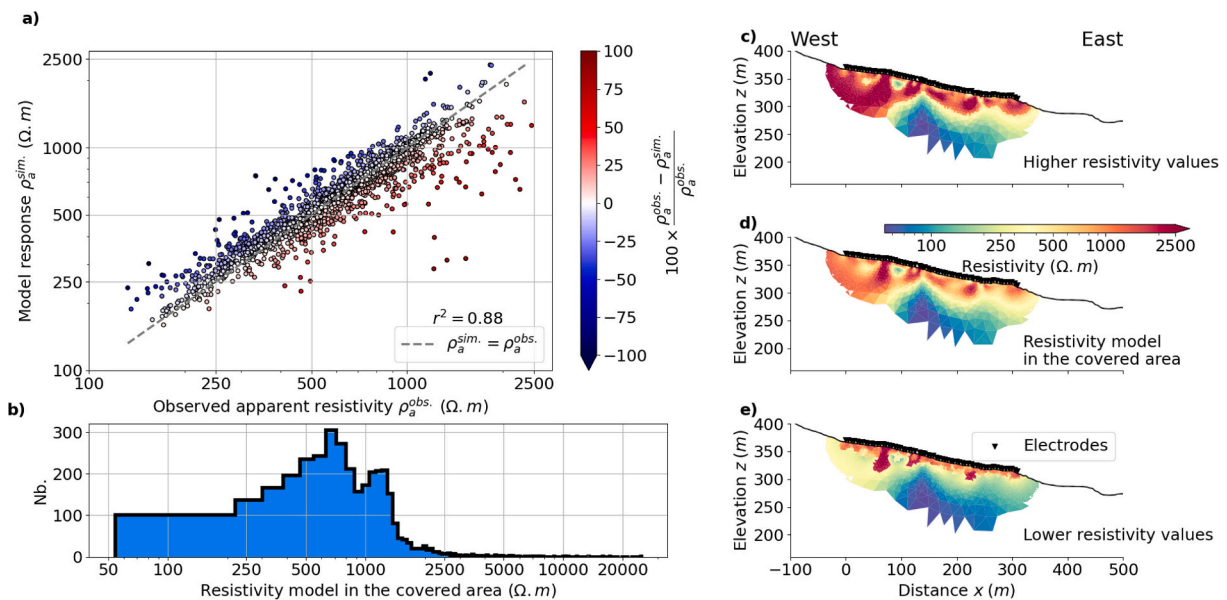


Fig. 6. Electrical resistivity tomography results. a) Misfit between observed and simulated data in terms of apparent resistivity, with r^2 the statistical coefficient of determination. The black dashed line corresponds to the perfect fit. b) Distribution of the calculated resistivity. c) Highest resistivity values in the covered area computed by the PyMERRY code (Gautier et al., 2023). d) ERT cross-section: resistivity model in the covered area computed by data inversion with PyGIMLi (Rücker et al., 2017). e) Lowest resistivity values in the covered area computed by the PyMERRY code.

data reveals a high correlation, confirming that the model reliably reproduces the observations (Fig. 6a). However, the distribution of electrical resistivity in the computed ERT model shows a few cells with particularly high resistivity cells, ranging from 2500 to 25,000 $\Omega\cdot\text{m}$ (Fig. 6b). Subsequently, we applied the PyMERRY software to the resulting 2D resistivity section to analyze the reliability of the ERT model (Gautier et al., 2023). This post-processing tool computes a coverage mask that defines the domain well-constrained by the data and the inversion process. Additionally, it provides the minimum and maximum resistivity values for each covered cell in the ERT image. These maps (Fig. 6c and e) reveal the largest errors the ERT model (Fig. 6d) may include. Together, these two outputs allow us to delineate the investigated zones, identify lateral variations in coverage, and quantify uncertainties in the resistivity model within the resolved area.

The mask obtained with PyMERRY gives a maximum investigation depth of ca. 100 m. It also underlines coverage differences between the two ends of the model due to west-east resistivity variations (Fig. 6c, d, and e). In addition to topography-related effects, the higher the resistivity, the less the area is correctly covered, leading to a westward decrease in the coverage. The covered resistivity model ranges from 50 to 25,000 $\Omega\cdot\text{m}$ (Fig. 6b), with the major resistivity distribution from 50 to 2500 $\Omega\cdot\text{m}$ and a median of 600 $\Omega\cdot\text{m}$. The ERT image derived from inversion (Fig. 6d) and the two sections associated with lower and higher resistivity values (Fig. 6c and e) show patches with aberrant resistivity values exceeding 10,000 Ω . PyMERRY allows us to better locate these patches and not interpret them in terms of lithology. These artefacts are located at horizontal distances of 70 m, 120 m, and 225 m from the westernmost electrode and may be associated with electrodes that exhibit poor coupling to the ground.

Despite these artefacts, all three ERT sections reveal the main features of the internal landslide structure, which include:

1. A shallow layer with resistivity values of ca. 1500 $\Omega\cdot\text{m}$, extending from the surface to a depth of 10 m.
2. An intermediate layer with an average thickness of 50 m and resistivity values varying from 300 $\Omega\cdot\text{m}$ to 1500 $\Omega\cdot\text{m}$. This layer shows sharp edges pointing upward at 130 and 230 m.

3. A deep layer with low resistivity values ranging from 50 to 300 $\Omega\cdot\text{m}$. ERT models do not provide a sharp image of the interface between the two deepest layers.

4.2. Modelling of density contrast

It is well known that the greater the number of mass bodies in a model, the better the fit to gravimetric data. However, the strong nonuniqueness of gravity inversion requires developing a parsimonious approach that balances the model's goodness-of-fit with its simplicity (Akaike, 1974). To this end, we do not invert the gravimetric data but employ direct modelling using a single mass body with a density contrast that represents the variation between colluvial deposits and the surrounding rocks. This body's geometry and density contrast are estimated by trial and error, using the chi-squared value as the consistency criterion between modeled and obtained Bouguer anomalies. Based on field observations and borehole information (Denchik et al., 2019), we assume that Bouguer anomalies are related to a colluvial deposit layer approximately 30 m thick. Hence, the approach consists of exploring the body's density contrasts between $-400 \text{ kg}\cdot\text{m}^{-3}$ and $-1000 \text{ kg}\cdot\text{m}^{-3}$ and lateral thickness variations ranging from 15 m to 50 m.

Our models, while simple, effectively explain the key features of the collected microgravimetric data. The best-fitting model, with a chi-squared value of 1.05, reveals the presence of an anomalous body within the landslide zone (Fig. 7). This body measures 700 m in length and 30 m in thickness, with a density contrast of $-650 \text{ kg}\cdot\text{m}^{-3}$. This simple geometry accounts for the long wavelength of the gravity signal observed between the Larzac plateau and the landslide. To better fit the shorter wavelength and account for slight fluctuations in gravity data within the landslide area, variations in the thickness of the anomalous body are necessary. Specifically, the thickness increases from 17 m near the Lergue river to 45 m below the westernmost electrode. Regarding the amplitude, the estimated error of approximately 0.1 mGal for the Bouguer anomalies yields a potential range for the density contrast of $-520 \text{ kg}\cdot\text{m}^{-3}$ to $-730 \text{ kg}\cdot\text{m}^{-3}$ (Fig. 7a).

5. Discussion

The various approaches, including structural and geomorphological

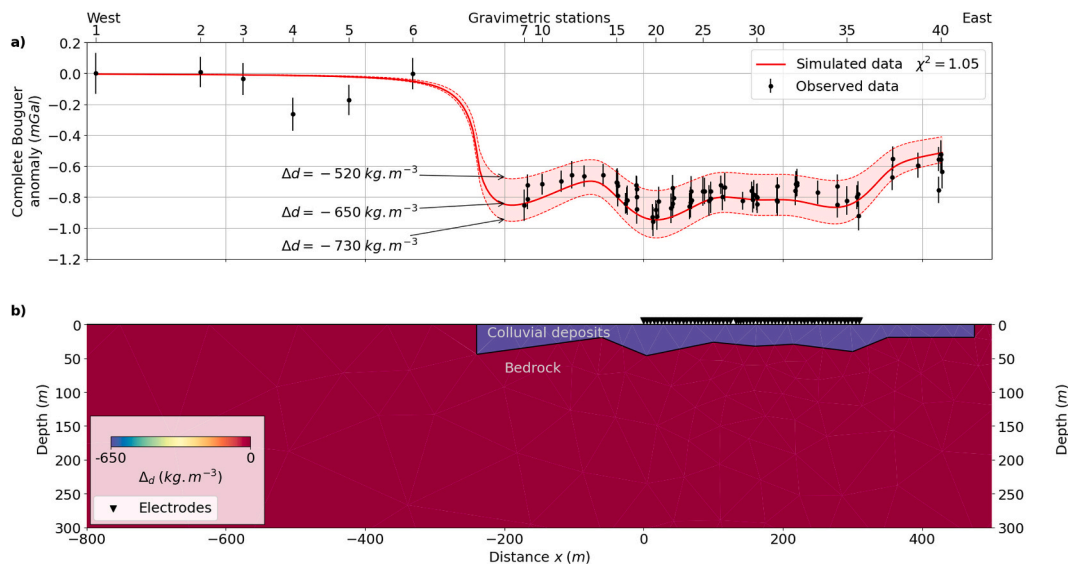


Fig. 7. Density contrast modelling. a) Observed (black circles) and simulated (red lines) gravimetric effects associated with the density model shown in b). χ^2 is the chi-squared value. The solid red line represents the complete Bouguer anomaly associated with a density contrast $\Delta d = -650 \text{ kg}\cdot\text{m}^{-3}$. Dashed red lines and red-filled area represent simulated anomalies associated with density contrasts ranging from $-730 \text{ kg}\cdot\text{m}^{-3}$ to $-520 \text{ kg}\cdot\text{m}^{-3}$. b) Density contrast model between limestone bedrock and colluvial deposits. The black triangles marked on the model surface indicate the locations of the electrical resistivity profile, shown as reference. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

observations, borehole measurements, electrical resistivity data, and gravimetric anomalies, presented in the previous sections, were conducted separately. However, they complement each other and together help to validate the results and support a coupled interpretation.

5.1. Complementarity of approaches

The geometry of the colluvial deposit, as inferred from microgravity data, aligns well with the projection of borehole observations in PEG1-GM and DEO onto the profile (Fig. 8), confirming the reliability of the microgravimetry to map lithological heterogeneities, as suggested by Del Gaudio et al. (2000). The gravimetry modelling supports substantial thickness variations, ranging from 45 m below the westernmost electrodes to 32 m near the middle of the electrical profile (Fig. 8). In contrast, the agreement between gravity results and observations in PEG1-BRGM is less reliable. The gravimetric approach suggests a thickness of 27 m, while borehole data indicate a colluvium base at 49 m. This discrepancy may be attributed to the location of the PEG1-BRGM borehole, approximately 155 m south of the gravimetric profile, whereas the PEG1-GM and DEO boreholes are within a few dozen meters of it (see map on Fig. 2). This spatial offset could account for the observed inconsistency, as lateral variations in colluvial deposit thickness may be present. This finding emphasizes the relevance and complementary nature of microgravimetric measurements and borehole observations, paving the way for future studies focused on characterizing lithological heterogeneities within the landslide in three dimensions.

As proposed by Ehrlich et al. (2018) and Sun et al. (2024), our ERT results (Fig. 8) confirm the reliability of this geophysical approach for imaging lithological heterogeneities in landslide contexts including colluvium materials within the near-surface. The joint analysis of the ERT image and the body geometry obtained from microgravimetry indicates that the colluvial deposit is associated with a zone exhibiting electrical resistivity greater than 300 $\Omega\cdot\text{m}$ (Fig. 8). This zone corresponds to the superficial and intermediate layers visible in the ERT image. Beneath these layers, resistivities ranging from 50 to 300 $\Omega\cdot\text{m}$ are related to limestone and marl lithologies. The resolution of the ERT and gravimetric images, as well as the contrast between electrical resistivity and density, is not sufficient to identify the massive Triassic evaporite layer. This layer and the damage zone at 45 m depth can only be accurately determined through borehole logs.

ERT offers relatively good resolution and an inversion process that

yields a well-resolved subsurface image, but with a limited depth of investigation. In comparison, gravimetric approaches have a greater depth of investigation and can be useful on sites with density contrasts. However, gravimetric data processing involves forward modelling of the gravimetric signal. This approach requires prior knowledge of the structures for which we want to assess the gravimetric effect, including the number, locations, and geometry of modeled bodies. Additionally, the non-uniqueness of the solution requires applying the principle of parsimony (Akaike, 1974). Our study demonstrates that, despite their inherent limitations, ERT and gravimetry, when constrained by borehole data, provide a robust and coherent characterization of subsurface heterogeneities.

5.2. Interpretative cross-section

Based on our results, available high-resolution DEM and previous geological studies (Denchik et al., 2019), we propose an interpretative cross-section that captures the main geometric features of the Pégairolles-de-l'Escalette landslide along the study profile (Fig. 9b).

In Fig. 8, the 300 $\Omega\cdot\text{m}$ contour line separates the top resistivity layer (interpreted as colluvium) and the low resistivity body (associated with marl). This contour line displays sharp edges at 130 m and 230 m, indicating localized variations in colluvial deposit thickness that we associated with the presence of tilted blocks. Supporting evidence from microgravity measurements reinforces this interpretation. Considering the relatively constant surface slope across the landslide, we attribute the short-wavelength gravity signal to abrupt changes in the geometry of the colluvium-limestone interface. Furthermore, as already mentioned by Denchik et al. (2019), our geological field observations confirm the presence of tilted limestone blocks at the surface, which we link to the overall fabric of the slide units. These units show a systematic tilting towards the footwall of the main active failure zone, consistent with a listric geometry.

Additionally, within 100 m west of the Lergue River, the topographic profile exhibits a sharp decrease in slope and a horizontal flattening of the surface. Gravimetric data further suggest the presence of a low-density body approximately 15 m thick in this area, which we interpret as colluvial and alluvial deposits. Lastly, we propose the existence of multiple imbricated sliding surfaces, associated with steep normal faulting at the landslide's upper extent (Elkharrat et al., 2024a and 2024b), horizontalizing towards the middle, and ultimately reaching the current elevation of the Lergue river. While this last interpretation is

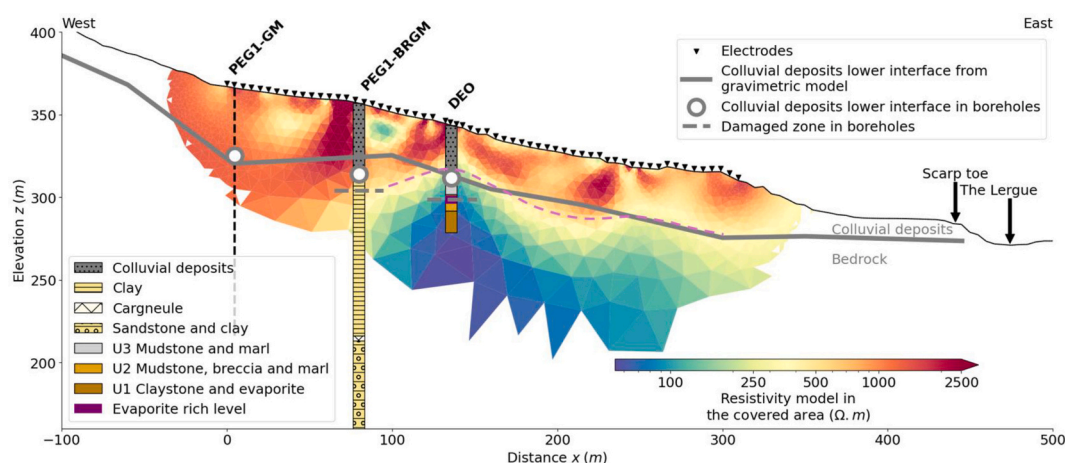


Fig. 8. Geophysical cross-section obtained using the implemented multi-method geophysical approach. Colorscale is associated with the obtained electrical resistivity tomography image. The thick grey line corresponds to the colluvium-limestone interface geometry inferred from microgravimetric data. White circles show the depth of the colluvium-limestone interface as observed in the PEG1-GM, PEG1-BRGM and DEO boreholes. The two rock columns indicate the main lithology found in borehole cuttings of PEG1-BRGM and DEO. Black dotted line represents the location of the PEG1-GM borehole. The pink dotted line represents a resistivity value of c.a. 300 $\Omega\cdot\text{m}$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

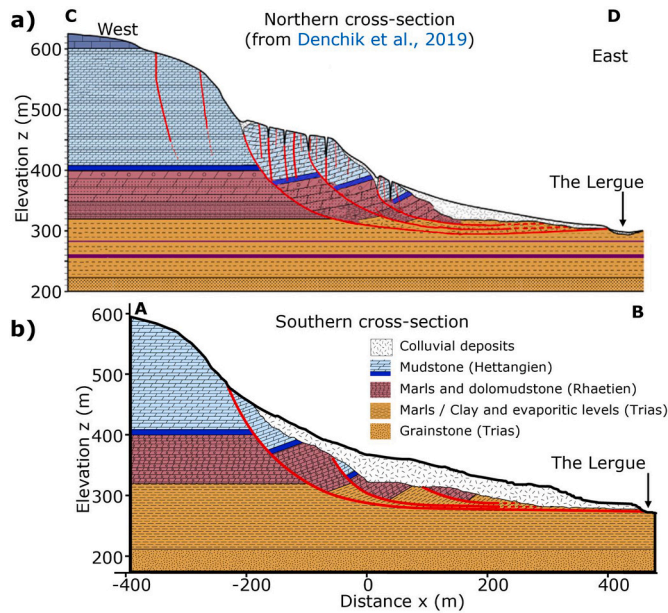


Fig. 9. Geological interpretative cross-section at the slope scale. a) Northern cross-section proposed by Denchik et al. (2019) from geological and one borehole observations (see location of the labelled profile C–D in Fig. 2). b) Southern cross-section of the central part of the landslide, proposed in this study from a multi-method approach that includes geomorphological observations, data from three boreholes, an electrical resistivity profile and micro-gravity measurements (refer to the location of the labelled profile A–B in Fig. 2).

somewhat limited by data constraints, it provides a plausible explanation for the observed features.

Our proposed interpretative cross-section (Fig. 9b) supports the rotational nature of the landslide, in agreement with the cross-section (Fig. 9a) presented by Denchik et al. (2019) along a transect slightly farther north (see location in Fig. 2). However, a comparative analysis of these two sections reveals significant north-south variations. These include tilted blocks observed at the outcrop and on the digital elevation model, as well as a reduction in colluvial deposits along the northernmost profile (Figs. 8 and 9). These discrepancies between the two sections may reflect limitations of the available data, as the Denchik et al. (2019) section is constrained in depth by a single borehole. Alternatively, these variations may indicate a complex three-dimensional geometry, which will be explored in future geophysical and geomorphological investigations across the entire slope.

5.3. Groundwater storage and flow

Because colluvial deposits are unconsolidated sediment, they contain a large volume of voids that can fill with water during heavy rainfall. Hence, colluvium plays a significant role in hydrological processes such as rainfall infiltration, ground saturation, or increased pore-water pressure, which control landslide stability.

Based on our results, we assess the effective porosity φ of the colluvial deposit, assuming that during our microgravimetric campaigns, this unit is perfectly dry (100% air-saturated). In that case, the density d of the porous medium is

$$d = (1 - \varphi) \times d_{\text{matrix}} + \varphi \times d_{\text{air}} \simeq (1 - \varphi) \times d_{\text{matrix}} \quad (1)$$

where $d_{\text{matrix}} = 2700 \text{ kg.m}^{-3}$ is the limestone density, and the air density d_{air} is assumed to be negligible. The average density contrast $\Delta d = d - d_{\text{matrix}} = -625 \text{ kg.m}^{-3}$ estimated from the gravity model can then be used to assess an effective porosity

$$\varphi = \frac{-\Delta d}{d_{\text{matrix}}} \quad (2)$$

and its associated uncertainties

$$\sigma_{\varphi} = \frac{1}{d_{\text{matrix}}} \times \sigma_{\Delta d} + \frac{-\Delta d}{d_{\text{matrix}}^2} \times \sigma_{d_{\text{matrix}}} \quad (3)$$

with $\sigma_{\Delta d} = 105 \text{ kg.m}^{-3}$ and $\sigma_{d_{\text{matrix}}} = 100 \text{ kg.m}^{-3}$, the uncertainties in the density contrast and the matrix density, respectively.

The effective porosity obtained is 0.23 ± 0.05 , indicating a high storage capacity and significant pore-water pressure variations due to heavy rainfall. Assuming a 36-m-high colluvium column 100% saturated with water, our findings indicate pressure variations of 0.81 ± 0.18 bar between the dry and wet seasons. This value is consistent with the ca. 1 ± 0.2 bar pressure variation observed in the WB borehole between 2012 and 2017 (see Fig. 5 in Denchik et al., 2019). It suggests a high degree of vertical infiltration and percolation within the colluvium, although we cannot rule out lateral groundwater flow.

6. Conclusion

This study demonstrates the value of a multi-method approach for characterizing the internal structure and water storage capacity of a deep-seated slow-moving landslide. By coupling electrical resistivity measurements, microgravimetry data acquisition, geological observations, and borehole data, we successfully described the internal structure of the landslide. Our approach yielded novel insights, particularly in delineating the geometry and effective porosity of the landslide's colluvial deposit.

In agreement with the study by Denchik et al. (2019), our proposed interpretative geological section supports the rotational nature of the landslide, as evidenced by surface-tilted limestone blocks and abrupt changes in the geometry of the colluvium-limestone interface. However, a detailed comparative analysis of the two sections reveals significant north-south variations, suggesting lateral variations in the landslide's structure that need to be documented in future studies.

Our findings reveal a high-resistivity layer ($> 300 \Omega.m$) with a negative density contrast of ca. -650 kg.m^{-3} . This layer spans over 700 m and has an average thickness of 30 m with significant thickness variations (15–50 m) attributed to geometric changes at the colluvium-bedrock interface, likely due to tilted limestone blocks. The obtained density contrast allowed us to estimate an effective porosity of 0.23 ± 0.05 , consistent with the pressure variations observed at 36 m depth between the dry and wet seasons.

Beyond the Pégairolles-de-l'Escalette landslide, our results emphasize the relevance of integrating conventional near-surface and borehole data with gravimetry to link subsurface geometry and materials, effective porosity, and rainfall-induced pore-pressure variations. This information is essential for a better understanding of slope stability and for improving landslide hazard assessments. The results demonstrate that gravimetry, although still underused in landslide investigation, can significantly enhance the characterization of deep internal structures and water storage when sufficient density contrasts are present. This opens new perspectives for applying such integrated approaches to other landslides, particularly where conventional methods alone fail to resolve deep-seated features and for advancing the monitoring of groundwater storage and flow dynamics within the slope.

Future work will focus on the landslide's internal 3D structure and its potential effects on both slope stability and surface deformation. While the geophysical-coupled approach proposed in this study is well-suited, new data acquisition will be needed to enhance spatial coverage, using the same methods or integrating new ones, such as ambient-noise seismic recording. Multimethods monitoring including gravimetry, seismic ambient noise methods and the commonly used time-lapse ERT will also be deployed to infer the landslide's dynamic, in particular its

response to heavy-rainfall seasons.

CRedit authorship contribution statement

Maxime Gautier: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis. **Stéphanie Gautier:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition. **Rodolphe Cattin:** Writing – review & editing, Validation, Supervision, Methodology, Investigation.

Funding sources

M. Gautier's Ph.D. is supported by a French Ministry for Higher Education and Research fellowship. This work has been supported by the OREME Observatory (Observatoire de REcherche Méditerranéen de l'Environnement) and the OMIV observatory (Observatoire des Mouvements et Instabilités de Versant, INSU, CNRS).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

We acknowledge valuable input from three anonymous referees. We thank the OREME Observatory (Observatoire de REcherche Méditerranéen de l'Environnement) and the OMIV observatory (Observatoire des Mouvements et Instabilités de Versants) for the permanent electrical resistivity tomography, borehole, and downhole monitoring data, and the petrophysical data (<https://dx.doi.org/10.25666/DATAOSU-2021-03-16>). The measurements were carried out with instruments of the Parc d'Instruments National PGravi of CNRS-INSU with the support of the Action Spécifique Gravimétrie of Epos-Fr for which we would like to thank Nicolas Le Moigne. These field campaigns would not have been so efficient without the researchers, interns, and students of Geosciences Montpellier's IGS-Team. Finally, we thank Matthieu Ferry for DEM processing and his expertise in the geomorphological features of the studied landslide.

Data availability

Data will be made available on request.

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